Aktivitas 10

Konsep Basis Data Terdistribusi

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Basis Data Terdistribusi

Konsep

Ketika sebuah organisasi tersebar di beberapa tempat, organisasi akan memilih menyimpan basis data secara terpusat atau secara terdistribusi. Basis data terdistribusi adalah sebuah basis data tunggal yang tersebar secara fisik di seluruh komputer di beberapa lokasi yang dihubungkan oleh jaringan komunikasi data

Alasan Penerapan Basis Data Terdistribusi

1. Unit Bisnis yang Terdistribusi
2. Berbagi Data
3. Biaya dan Keandalan Komunikasi Data
4. Aplikasi dari Beberapa Pengembang (Vendor)
5. Database Recovery
6. Sinkronisasi Data

Opsi Pada Basis Data Terdistribusi

1. Replikasi Data (*Data Replication*)
2. Partisi Horizontal (*Horizontal Partitioning*)
3. Partisi Vertikal (*Vertical Partitioning*)
4. Kombinasi (*Combinations*)

Strategi Basis Data Terdistribusi

Strategy	Reliability	Expandability	Communications Overhead	Manageability	Data Consistency
Centralized	**POOR:** Highly dependent on central server	**POOR:** Limitations are barriers to performance	**VERY HIGH:** High traffic to one site	**VERY GOOD:** One monolithic site requires little coordination	**EXCELLENT:** All users always have the same data
Replicated with snapshots	**GOOD:** Redundancy and tolerated delays	**VERY GOOD:** Cost of additional copies may be less than linear	**LOW to MEDIUM:** Not constant, but periodic snapshots can cause bursts of network traffic	**VERY GOOD:** Each copy is like every other one	**MEDIUM:** Fine as long as delays are tolerated by business needs
Synchronized replication	**EXCELLENT:** Redundancy and minimal delays	**VERY GOOD:** Cost of additional copies may be low and synchronization work only linear	**MEDIUM:** Message are constant, but some delays are tolerated	**MEDIUM:** Collisions add some complexity to manageability	**MEDIUM to VERY GOOD:** Close to precise consistency
Integrated Partitions	**VERY GOOD:** Effective use of partitioning and redundancy	**VERY GOOD:** New nodes get only data they need without changes in overall database design	**LOW to MEDIUM:** Most queries are local, but queries that require data from multiple sites can cause a temporary load	**DIFFICULT:** Especially difficult for queries that need data from distributed tables, and updates must be tightly coordinated	**VERY POOR:** Considerable effort; and inconsistencies not tolerated
Decentralized with independent partitions	**GOOD:** Depends on only local database availability	**GOOD:** New sites independent of existing ones	**LOW:** Little if any need to pass data or queries across the network (if one exists)	**VERY GOOD:** Easy for each site, until there is a need to share data across sites	**LOW:** No guarantees of consistency: in fact, pretty sure of inconsistency

Selamat Belajar...

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